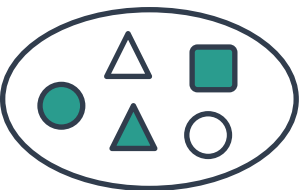


Dataset

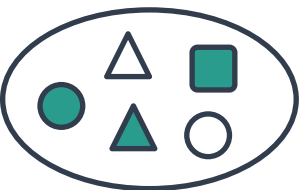


cardinality
operator



F_0

(ϵ, δ) -approximation



Sketch



$\hat{F}_0$

where (ϵ, δ) -approximation means

$$\forall \epsilon \exists \delta : \mathbb{P}[|F_0 - \hat{F}_0| > \epsilon F_0] < \delta$$